Protein purification by IMAC (Ni-NTA)

What you're going to get / What you expect

Purified His-tagged enzyme, dialysed and concentrated, ready for characterisation.

Materials list (per 100 mL culture)

- Induced biomass from 100 mL of culture.
- Resuspension buffer: Tris-HCl 50 mM pH 8.0, NaCl 300 mM, imidazole 20 mM.
- Elution buffer: resuspension buffer with 500 mM imidazole.
- Dialysis buffer: Tris-HCl 10 mM pH 7.0, NaCl 100 mM.
- Benzonase endonuclease; Ni-NTA agarose resin.
- Pressure homogeniser (e.g. PandaPLUS 2000).
- Dialysis membrane MWCO 3.5 kDa; ultrafiltration concentrator MWCO 10 kDa.

Steps

1. Split the biomass into two 50 mL fractions and collect by centrifugation at 3,500 x g for 20 minutes at 10 ºC.
2. Resuspend each pellet in 20 mL of resuspension buffer at 4 ºC and add 5 uL of benzonase to reduce lysate viscosity.
3. Disrupt by pressure homogenisation at 4 ºC; clarify by high-speed centrifugation at 14,600 x g for 30 minutes at 4 ºC and keep the supernatant (Fraction 1).
4. Equilibrate 1.6 mL of Ni-NTA resin with resuspension buffer, add the clarified supernatant and incubate 1 hour at 4 C under gentle agitation.
5. Centrifuge at 3,500 x g for 15 minutes at 4 ºC and keep the supernatant as the non-retained fraction (Fraction 2).
6. Wash the resin with 20 mL of resuspension buffer (20 mM imidazole) for 10 minutes at 4 ºC and sample it (Fraction 3). The 20 mM imidazole displaces non-specific binders without eluting the tagged enzyme.
7. Elute with 5 mL of elution buffer (500 mM imidazole) (Fraction 4).
8. Dialyse overnight against 2 L of dialysis buffer (MWCO 3.5 kDa), twice (Fraction 5).
9. Concentrate by ultrafiltration (MWCO 10 kDa) at 3,500 x g until the volume is reduced by 90% (Fraction 6); monitor every stage by SDS-PAGE.

Errores comunes / troubleshooting

- Faint elution bands or low yield usually track low soluble expression upstream; the tag chemistry still works, so enough protein for characterisation can still be recovered.
- Non-specific contaminants co-eluting: the 20 mM imidazole in the binding and wash buffers should suppress them; raise the wash imidazole slightly if needed.
- Use the two membrane cut-offs as intended: 3.5 kDa to dialyse out small solutes, 10 kDa to retain the enzyme during concentration.